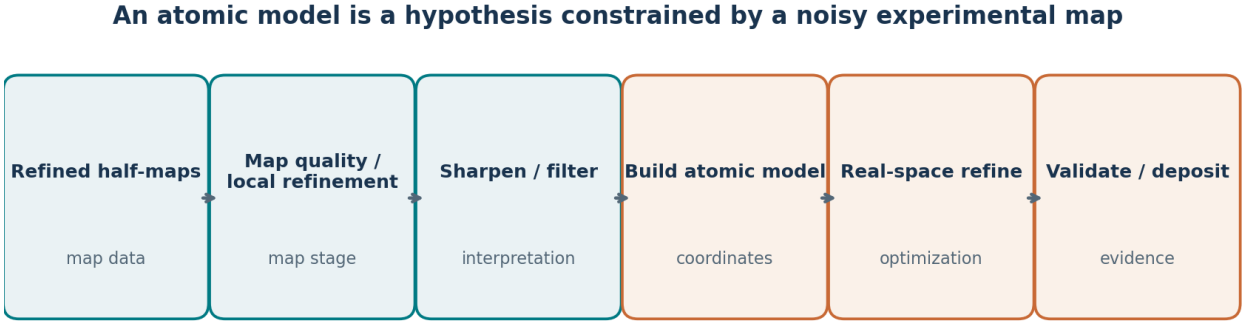
A MATHEMATICAL INTRODUCTION TO SINGLE-PARTICLE CRYO-EM

From a 3D Map to an Atomic Model

Map Postprocessing, Model Building, Real-Space Refinement & Validation

A textbook chapter with first principles, derivations, original diagrams, numerical examples, exercises, a practical workflow, and linked primary literature



Chapter III • Companion to the chapters on imaging/CTF and reconstruction/classification

28 October 2026 | CryoSPARC, Phenix, ModelAngelo, Coot, ChimeraX, ISOLDE, validation

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How to read this chapter

Chapter I derived electron scattering, defocus and the contrast transfer function. Chapter II derived particle reconstruction, orientation estimation, statistical classification, PCA and conformational variability. This chapter begins at their output: a reconstructed three-dimensional scalar field, often distributed as an MRC/CCP4 map with two independently refined half-maps. It explains the remaining scientific steps required to defend an atomic structural interpretation. No previous familiarity with crystallographic model refinement or molecular mechanics is assumed. This is not an operating procedure specific to any software version; program descriptions refer to official documentation cited in the text.

Key concept

Two very different meanings of refinement. In particle refinement, the input is noisy 2D particles and the output is an improved 3D map, poses and related uncertainty. In *atomic real-space refinement*, the input is a 3D map plus a proposed atomic model; the output is an adjusted atomic model. No atomic refinement method can create experimentally supported detail that is not present in the map.

1 What remains after reconstruction and classification?

The workflow graphic motivating this chapter correctly identifies the conventional stages: micrographs, motion/CTF correction, particle picking, 2D classification, 3D reconstruction, 3D classification/refinement, 3D density map, atomic model building, real-space refinement, and validation. This is a *conceptual dependency graph*, not a mandatory one-way pipeline: classes may be re-refined, local regions may need separate reconstructions, and atomic modeling repeatedly cycles through refinement and validation. Some projects end with a validated map because the resolution does not justify atomic coordinates. These steps are based on

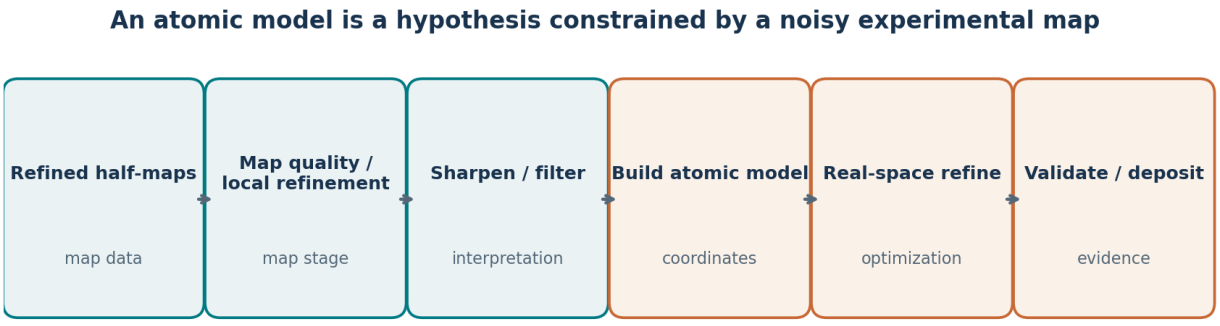


Figure 1: Two linked downstream levels. The first three operations assess and improve the reconstructed map; the latter three interpret that map as an atomic model. A low-resolution map may never reach the coordinate stage.

official CryoSPARC, Phenix and EMDB workflows [1, 4, 9, 20].

2 Prerequisite: what is a reconstructed 3D map?

2.1 A scalar field, not a list of atom positions

A 3D map is a function $M : \mathbb{R}^3 \to \mathbb{R}$ evaluated on a finite three-dimensional grid. At a voxel center $\boldsymbol{r}_{ijk}$ the stored number M_{ijk} is a reconstruction-derived map value. Cryo-EM traditionally calls it a "density map," but

Stage	Question the stage answers
Independent half-map checks	Do independent particle halves support the claimed frequencies?
Local resolution / anisotropy	Is every part and direction equally interpretable?
Map filtering and sharpening	How should the noisy Fourier information be displayed for tracing?
Local refinement / subtraction	Can a flexible domain be reconstructed more clearly?
Model fitting / building	Which protein, nucleic-acid and ligand coordinates explain the map?
Real-space atomic refinement	What conformations fit the map without violating chemistry?
Independent validation / deposition	Which claims survive out-of-sample and stereochemical checks?

for elastic electron imaging its contrast more directly reflects specimen electrostatic potential (after an entire imaging and reconstruction pipeline), not the X-ray electron-density distribution in the literal crystallographic sense.

$$M_{ijk} \approx (h * V)(\boldsymbol{r}_{ijk}) + \epsilon_{ijk}, \qquad \boldsymbol{r}_{ijk} = \boldsymbol{r}_0 + a(i,j,k)^T. \tag{1}$$

V is an underlying specimen potential, h an effective reconstruction point-spread function, a the voxel spacing and ϵ the reconstruction error. This is a pedagogical approximation: actual reconstructions have nonuniform, correlated and often anisotropic errors.

Key concept

Voxel versus atom. A voxel is a sample of a continuous field, not a physical cube occupied by one atom. An atom is a chemically identified nucleus with element, residue and 3D coordinates. At sufficiently high resolution, a pattern of map peaks constrains atom positions; at lower resolution, the same map may only support domains or helices.

2.2 Grid coordinates and physical coordinates

An MRC map stores voxel dimensions, grid geometry and an origin or start offset. A coordinate model stores atomic positions in angstroms. They must share the same origin, handedness and orientation before fitting. The mapping is conceptually $\boldsymbol{r} = \boldsymbol{r}_0 + A\boldsymbol{n}$ for voxel index $\boldsymbol{n} = (i,j,k)^T$, where A is the coordinate basis (usually aI for a cubic grid). Arbitrarily rotating or recentering one without updating the other produces apparently terrible map–model correlation even when the structure is correct. EMDB specifically checks map/model coordinate consistency [20].

2.3 Contours are visualization, not an isosurface of matter

A contour surface plots $\{\boldsymbol{r} : M(\boldsymbol{r}) = \tau\}$ at an investigator-chosen threshold τ. Changing τ changes which regions appear solid, but it does not change the underlying map or prove that a weak feature is absent. When comparing two conditions, use comparable normalization, filtering and contours; otherwise a purely display-induced difference can look like a real structural change.

2.4 The mathematically unavoidable effect of finite resolution

If $\widehat{V}(\boldsymbol{q})$ is the ideal transform, a filtered reconstruction may have $\widehat{M}(\boldsymbol{q}) = H(\boldsymbol{q})\widehat{V}(\boldsymbol{q}) + \widehat{\epsilon}(\boldsymbol{q})$. For an isotropic Gaussian blur of standard deviation σ, $H(\boldsymbol{q}) = \exp(-2\pi^2\sigma^2\|\boldsymbol{q}\|^2)$. High spatial frequencies are attenuated first. Two nearby atoms can merge into one continuous blob even if their underlying locations are distinct. This is the basic *identifiability* limitation of atomic fitting.

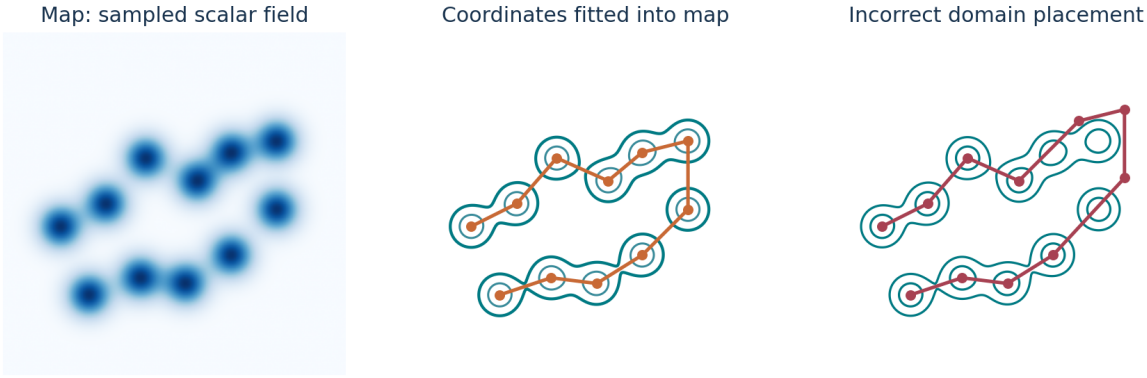


Figure 2: Original synthetic two-dimensional map and coordinate-model analogy. A contour is not an atom; coordinates must be independently assigned. The intentionally misplaced domain illustrates a model that may look plausible globally but disagree locally.

Worked example

Sampling versus resolution. Consider a map sampled at voxel spacing $a = 1.2$ Å. The Nyquist limit is $q_N = 1/(2a) = 0.417$ Å^{-1}, corresponding to $d_N = 2.4$ Å. That is only a sampling bound. If the actual experimental FSC resolution is 4.5 Å, the map does *not* thereby contain 2.4 Å information. Oversampling cannot restore missing frequencies.

3 Independent half maps and Fourier-shell correlation

3.1 Why two maps?

Split particles into independent subsets and reconstruct M_1 and M_2 separately under a gold-standard processing strategy. A simplified shell-wise model is $\widehat{M}_1 = S + N_1$ and $\widehat{M}_2 = S + N_2$, where S is shared signal and the random errors are independent. The Fourier-shell correlation (FSC) at spatial frequency q is

$$\mathrm{FSC}(q) = \frac{\sum_{\boldsymbol{q}\in\Omega_q} \widehat{M}_1(\boldsymbol{q})\widehat{M}_2^*(\boldsymbol{q})}{\sqrt{\sum_{\Omega_q} |\widehat{M}_1|^2 \sum_{\Omega_q} |\widehat{M}_2|^2}}, \tag{2}$$

where Ω_q denotes a thin Fourier shell; software commonly reports the real-valued shell correlation. An FSC curve reports agreement as a function of spatial frequency rather than one global image similarity score. A commonly used resolution estimate corresponds to the FSC 0.143 crossing in gold-standard half-map refinement [8, 6].

3.2 Why does 0.143 appear?

With equal independent half-map noise and a simplified signal-to-noise model, $r = \mathrm{SNR}_{\mathrm{half}}/(1 + \mathrm{SNR}_{\mathrm{half}})$. Combining the halves doubles their signal-to-noise ratio, giving $r_{\mathrm{full}} = 2r/(1 + r)$. At $r = 0.143$, $r_{\mathrm{full}} \approx 0.250$. The FSC threshold is an operational convention connected to SNR assumptions, not a guarantee that every atom is resolved at that spacing; masking and independence conditions matter.

3.3 Masking and false agreement

A tight hard mask multiplies both half maps by the same real-space window. Its Fourier transform convolves both maps, potentially creating correlated spectral artifacts and inflating apparent agreement. Use suitable

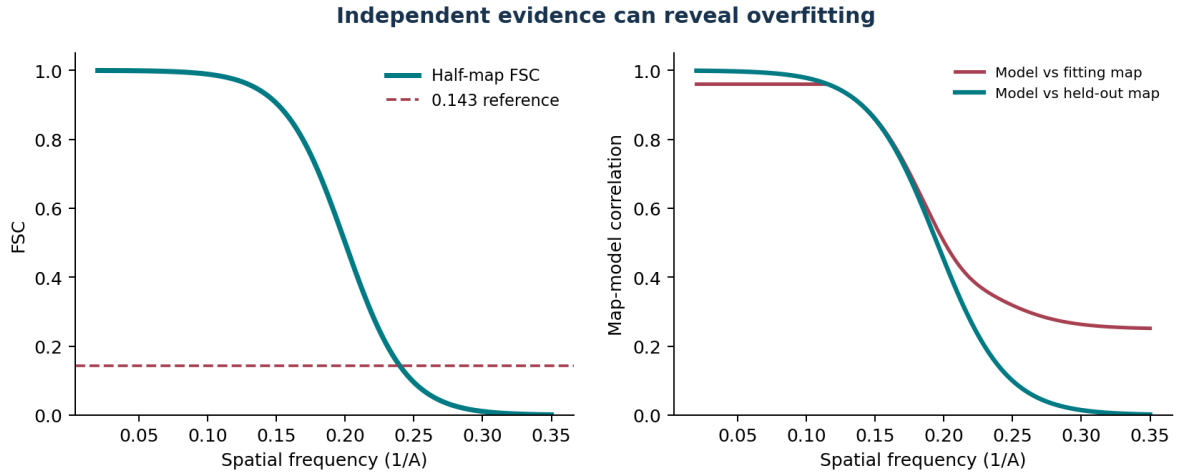


Figure 3: Schematic validation curves. The left panel illustrates half-map FSC and the commonly used 0.143 threshold. The right illustrates an overfit atomic model: it correlates more strongly with the map used for fitting than with an independent map. Curves are synthetic, not measured data.

soft edges; examine masked and unmasked curves; avoid tuning masks to obtain a desired resolution [6, 19]. Independent half-map processing also needs safeguards against reference leakage, overfitting and inadvertent correlation between subsets.

3.4 Local resolution versus global resolution

A protein with one rigid domain and a mobile loop may have, for example, 3.2 Å local resolution in the core and 6.5 Å in the loop. A single global FSC number cannot describe both. Local methods compute correlation within spatial windows, balancing Fourier localization and spatial locality; estimates depend on window size and other choices. CryoSPARC provides Local Resolution Estimation and local filtering [3, 4].

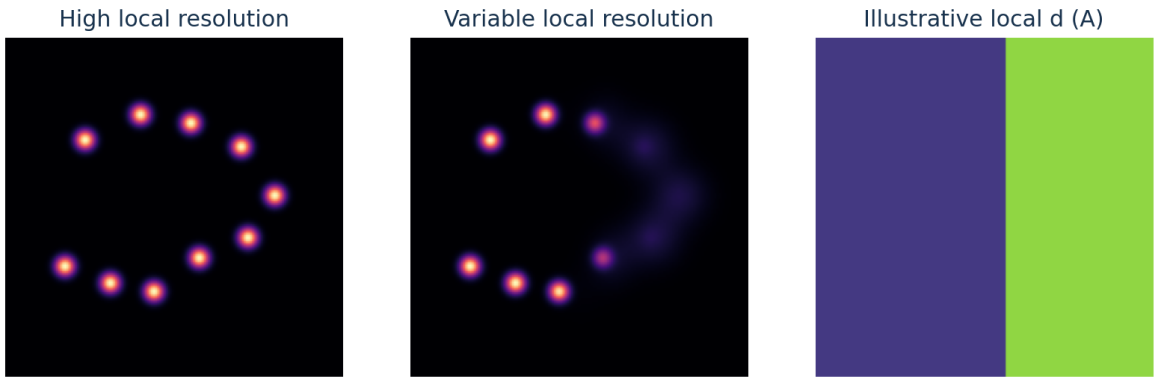


Figure 4: Synthetic high-resolution region versus spatially varying blur. The right panel is a conceptual local-resolution field, not the output of CryoSPARC. Local detail determines whether a coordinate model is defensible.

3.5 Directional anisotropy

When particle orientations are strongly biased, frequencies can be well measured in some directions and poorly measured in others. One may compute directional or conical FSCs, rather than only isotropic shells. CryoSPARC Orientation Diagnostics includes conical FSC-based measures and Fourier-sampling diagnostics [7]. For an elongated structure, an apparent helix displacement along a poorly sampled direction may be much less certain than its nominal global resolution suggests.

Worked example

Numerical FSC interpretation. Suppose the estimated half-map FSC at $q = 0.20\,\text{Å}^{-1}$ is $r = 0.60$. The equal-noise approximation gives $\text{SNR}_{\text{half}} = r/(1-r) = 1.5$, and the corresponding full-map FSC is $2r/(1+r) = 0.75$. The spatial frequency corresponds to $d = 5\,\text{Å}$. This is *not* proof of 5 Å resolution in every orientation or residue.

4 Map sharpening, filtering and masking

4.1 What is a B-factor envelope?

Many effects suppress high-frequency Fourier amplitudes. A convenient isotropic envelope is written

$$\widehat{M}_{\text{obs}}(s) \simeq \widehat{M}_{\text{ideal}}(s) \exp\left(-\frac{B_{\text{eff}} s^2}{4}\right) + \widehat{N}(s), \quad s = \|\boldsymbol{q}\|. \tag{3}$$

The global effective B_{eff} here parameterizes overall signal decay; it is not necessarily a molecular atom's thermal-displacement parameter. A linear fit of log-amplitude decay versus s^2 is often related to a Guinier-plot estimate. CryoSPARC sharpening tools allow choosing an explicit sharpening B-factor [5].

4.2 Sharpening is a Fourier multiplier

Apply $H_B(s) = \exp[-B_{\text{sharp}} s^2/4]$ to the map. Negative B_{sharp} increases high frequencies:

$$\boxed{\widehat{M}_{\text{sharp}}(\boldsymbol{q}) = \exp[-B_{\text{sharp}}\|\boldsymbol{q}\|^2/4]\widehat{M}_{\text{obs}}(\boldsymbol{q}).} \tag{4}$$

The operation can make side chains easier to see, but it amplifies noise at the same frequencies. Filtering or tapering suppresses frequencies beyond supported resolution. A locally filtered map adapts the cutoff to regional resolution [4].

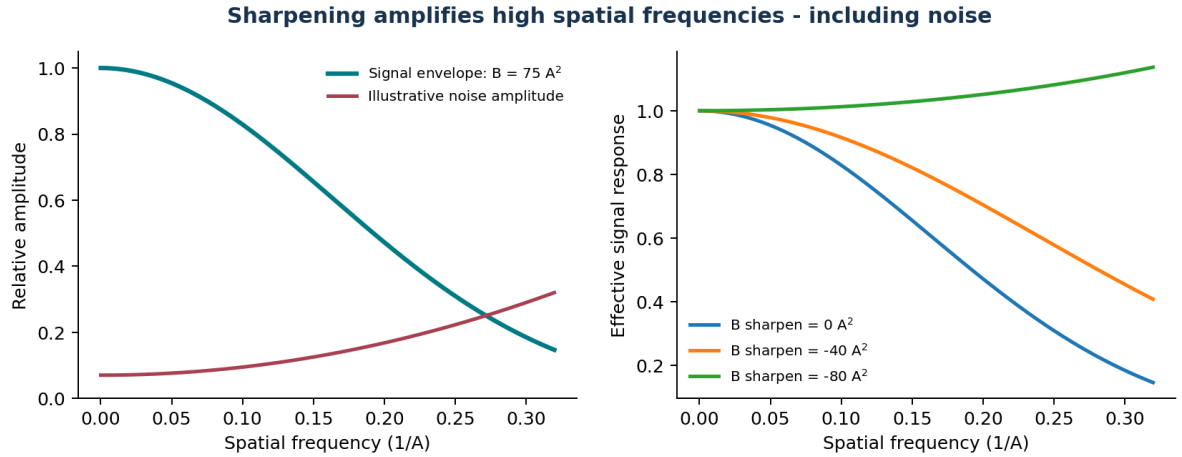


Figure 5: Illustrative envelope and sharpening curves. Negative sharpening B-factors compensate signal decay, but do not distinguish signal from noise. Curves are mathematical examples, not measurements.

Worked example

How much does sharpening amplify a 4 Å feature? A feature of period $d = 4\,\text{Å}$ has $s = 0.25\,\text{Å}^{-1}$. If $B_{\text{eff}} = 75\,\text{Å}^2$, its envelope is $e^{-75(0.25)^2/4} = e^{-1.172} = 0.310$. Sharpening with $B_{\text{sharp}} = -50\,\text{Å}^2$ multiplies

it by $e^{+50(0.25)^2/4} = 2.18$. The combined signal response becomes 0.676. The high-frequency noise is also amplified 2.18-fold.

4.3 Why maintain multiple map versions?

Keep the unsharpened/unfiltered half maps as primary evidence, the full map for fitting, and sharpened or locally filtered maps for human interpretation. Do not overwrite the independent half maps with a beautifully enhanced visualization. Model refinement methods vary in which map representation they prefer; for example, Servalcat describes using unsharpened maps in its specific refinement/statistical procedure [14]. The optimal display map is not necessarily the optimal refinement target.

Interpretation warning

Map enhancement cannot certify a feature. A neural postprocessor, strong sharpening, or denoising may make a plausible-looking side chain appear. The independent half maps and original reconstruction, not the aesthetic clarity of the displayed map, must support the structural interpretation. Any model used to enhance a map also introduces potential model bias.

5 Local refinement and particle subtraction

5.1 Why global refinement leaves some domains blurry

Suppose the complex is composed of two rigid domains, V_A and V_B, whose relative orientation fluctuates across particles. A global reconstruction estimates a common volume:

$$\bar{V}(\boldsymbol{r}) = \mathbb{E}_{R_B}\left[V_A(\boldsymbol{r}) + V_B(R_B^{-1}\boldsymbol{r})\right]. \tag{5}$$

If R_B varies, the average smears domain B. This can happen even if each individual copy of B has a well-defined internal structure. The result is ensemble averaging, not necessarily poor optics. *Local refinement* focuses pose optimization on a selected subvolume, often with a soft mask, to align a domain more consistently [1, 2].

5.2 What does a refinement mask do?

Let $W(\boldsymbol{r}) \in [0, 1]$ be a smooth spatial mask. The selected subvolume is $V_B^W = WV$. Mask selection defines the region that drives local alignment; it cannot remove the need for sufficient independent signal or broad angular coverage. Strong masks can introduce Fourier artifacts, so mask edges should be soft. Refined domain orientations may not correspond to one consistent rigid-body pose of the entire molecule.

5.3 Subtracting the rest of the molecule

In a simplified linear particle-image model,

$$y_i = A_i(V_A + V_B) + n_i. \tag{6}$$

If $\widehat{V}_A$ and its projection are accurate, subtract the predicted unwanted image signal:

$$y_i^{\text{sub}} = y_i - A_i\widehat{V}_A \approx A_iV_B + n_i + A_i(V_A - \widehat{V}_A). \tag{7}$$

The last term is *subtraction error*. Inaccurate pose, scale, masks or source maps can leave misleading positive/negative residuals. Subtraction can improve local alignment, but it never physically changes the specimen [1].

5.4 Why a local map is not automatically an atomic model

A focused high-resolution reconstruction might improve one domain but leave its relative placement uncertain. To interpret a whole complex, one needs a consistent reference frame and may require multi-body modeling or ensembles. Combining separately reconstructed, independently aligned domains into a single artificial rigid structure can erase real flexibility. Record both local-domain maps and the consensus assembly geometry.

Worked example

A one-pixel subtraction example. The measured signal is $y = 12$, with V_A contributing 9, V_B contributing 2, and noise 1. Subtracting a correct model $\widehat{V}_A = 9$ gives $y^{\text{sub}} = 3 = 2 + 1$. But subtracting an incorrect $\widehat{V}_A = 11$ gives $y^{\text{sub}} = 1 = 2 + 1 - 2$, misleadingly suppressing domain B. Subtraction depends on the accuracy of the *unwanted* signal estimate.

6 Prerequisite: chemistry and coordinate models

6.1 What is an atomic model?

A protein model comprises atoms indexed $j = 1, \ldots, N$ with positions $\boldsymbol{r}_j \in \mathbb{R}^3$, element types, residue identities, chain IDs, connectivity and possibly occupancies o_j and displacement parameters B_j. The coordinate vector has $3N$ components:

$$\boldsymbol{\theta} = (x_1, y_1, z_1, \ldots, x_N, y_N, z_N)^T. \tag{8}$$

An mmCIF model can describe chain sequences, ligands, alternate conformations and metadata in addition to coordinates. Atom coordinates are not automatically experimentally determined merely because they appear in a PDB/mmCIF file.

6.2 Bond lengths and bond angles

For bonded atoms i, j, the distance is

$$d_{ij} = \|\boldsymbol{r}_i - \boldsymbol{r}_j\|. \tag{9}$$

For atoms i–j–k, the bond angle is

$$\theta_{ijk} = \arccos \frac{(\boldsymbol{r}_i - \boldsymbol{r}_j) \cdot (\boldsymbol{r}_k - \boldsymbol{r}_j)}{d_{ij} d_{jk}}. \tag{10}$$

Covalent geometry is constrained by chemistry. The map alone rarely determines every bond or angle because many local shifts can preserve overall density fit.

6.3 Dihedral angles, rotamers and backbone conformations

A dihedral angle measures rotation about a bond connecting two planes. In proteins, backbone torsions ϕ and ψ describe local conformation, and side-chain torsions $\chi_1, \chi_2, \ldots$ describe rotamers. Steric and energetic constraints favor particular combinations. The empirical Ramachandran distribution $p(\phi, \psi \mid \text{residue type})$ is a strong validation reference, not an immutable physical law. Unusual conformations can be genuine if independently supported [17, 13].

6.4 What determines model complexity?

For N atoms, unconstrained coordinates have $3N$ degrees of freedom. Individual occupancies, alternate conformers and displacement parameters add more. At low resolution, experimental information may support rigid-body motion of helices or domains, but not thousands of independent atomic displacements. Appropriate restrictions on degrees of freedom are therefore an integral part of structural inference.

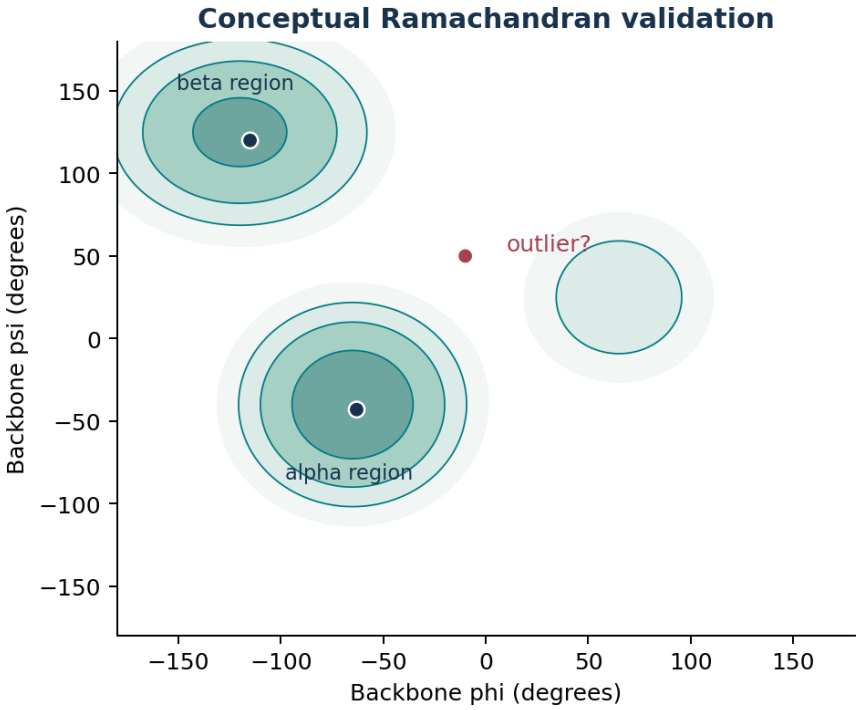


Figure 6: Conceptual Ramachandran plot showing illustrative allowed backbone regions and a putative outlier. This is a schematic Gaussian mixture, not the empirical MolProbity residue-specific distribution.

Key concept

Hierarchy of claims. At coarse resolution: overall envelope and domain architecture. At intermediate resolution: secondary-structure placement and perhaps backbone tracing. At higher local resolution: sequence registration, side-chain rotamers, ligand chemistry and individual atomic interactions. The numerical resolution cutoffs are context-dependent, not universal steps.

7 Initial model fitting: rigid bodies and coordinate alignment

7.1 Where does an initial model come from?

Common sources include experimental homolog structures, a known structure of the same protein in another state, AlphaFold-predicted coordinates, or a de novo map-traced model. The initial model is a *hypothesis*. Predicted-structure confidence measures confidence in the prediction, not independent experimental support for the fitted cryo-EM conformation. In a flexible protein, a high-confidence folded domain can still have an incorrect domain orientation [12, 11].

7.2 Rigid-body placement

A rigid-body transform maps each original atomic coordinate $\boldsymbol{x}_j$ to

$$\boldsymbol{r}_j = R\boldsymbol{x}_j + \boldsymbol{t}, \qquad R^T R = I, \quad \det R = 1. \tag{11}$$

The orientation $R \in SO(3)$ has three rotational degrees of freedom and $\boldsymbol{t}$ has three translational degrees. A rigid-body fit may maximize correlation between an atom-derived model map $M_c(R, \boldsymbol{t})$ and the experimental map M_o:

$$(\widehat{R}, \widehat{\boldsymbol{t}}) = \arg\max_{R,\boldsymbol{t}} \mathrm{CC}(M_o, M_c(R, \boldsymbol{t})). \tag{12}$$

Tools such as ChimeraX perform map visualization and rigid-body fitting; domain-wise fitting is often preferable to forcing a flexible assembly to act rigidly [12].

7.3 The Kabsch alignment problem

When corresponding points are known, a standard alignment minimizes weighted squared distances:

$$\min_{R\in SO(3),\boldsymbol{t}} \sum_{j=1}^{n} w_j \|R\boldsymbol{x}_j + \boldsymbol{t} - \boldsymbol{y}_j\|^2. \tag{13}$$

After calculating weighted centroids $\bar{\boldsymbol{x}}$ and $\bar{\boldsymbol{y}}$, the optimal translation for a given R is $\boldsymbol{t} = \bar{\boldsymbol{y}} - R\bar{\boldsymbol{x}}$. Center both point sets, put centered rows into matrices X and Y, and define $H = X^T W Y$. Compute the singular-value decomposition

$$H = U\Sigma V^T, \tag{14}$$

where W is a diagonal matrix of weights. For the column-vector convention of Eq. (11),

$$\boxed{R = VDU^T, \qquad D = \operatorname{diag}\bigl(1, 1, \det(VU^T)\bigr).} \tag{15}$$

The determinant correction forbids a mirror reflection. The optimal root-mean-square deviation is

$$\mathrm{RMSD} = \sqrt{\frac{\sum_j w_j \|R\boldsymbol{x}_j + \boldsymbol{t} - \boldsymbol{y}_j\|^2}{\sum_j w_j}}. \tag{16}$$

This is a coordinate-to-coordinate alignment. Fitting coordinates to a *density map* is a related but different optimization because atom correspondences are not explicitly given by the map.

Worked example

A translation-only fit. Three model marker points have coordinates $(0,0,0)$, $(1,0,0)$, $(0,1,0)$ Å. Their matched target points are $(3,-2,1)$, $(4,-2,1)$, $(3,-1,1)$ Å. The centroids differ by $(3,-2,1)$ Å; the optimal rigid transform is $R = I$, $\boldsymbol{t} = (3,-2,1)$ Å, giving RMSD zero. If a rigid fit requires one domain to be rotated differently from another, that is evidence against a *single* global rigid transform.

7.4 Fitting by fragments rather than an entire chain

At intermediate resolution, one may first position stable helices, sheets or domains as rigid fragments and then repair loops and connecting residues. This reduces the number of independently adjustable parameters. The risk is that individually fitted fragments may create impossible chain continuity or steric collisions; these must be resolved in subsequent chemically restrained steps.

8 De novo building and automated tools

8.1 Backbone tracing

At sufficient resolution, continuous map features can trace a polypeptide backbone, with alpha helices and beta strands supplying geometric clues. Carbon alpha positions are separated by roughly 3.8 Å along an ordinary peptide backbone, but the actual coordinates are constrained by chemistry and map resolution. Building a chain requires identifying the connectivity, residue order, sequence registration and local handedness; drawing a smooth line through the map is insufficient.

8.2 Sequence registration

Suppose a backbone trace contains L residue positions but their identities are uncertain. One must assign a sequence segment and residue register to the trace. Side-chain shapes, aromatic residues, proline-induced turns, secondary structure and independently known chain architecture help discriminate registers. At inadequate local resolution, neighboring sequence assignments may fit almost equally well; the correct response is reduced confidence rather than forced certainty.

8.3 ModelAngelo as a concrete example

ModelAngelo combines map information, protein sequence, and structural priors in a neural model to construct protein and nucleotide models. In its published method, a CNN proposes residue positions, a graph neural network refines positions and orientations, and post-processing builds the atomic model [11]. It improves efficiency but does not eliminate manual inspection: the authors specifically caution about regions with local resolution worse than roughly 3.5–4 Å and recommend refinement and human verification.

8.4 Where Coot, ChimeraX and ISOLDE fit

Coot supports manual chain tracing, residue mutation, rotamer adjustment, local geometry and map fitting. ChimeraX visualizes maps and models, aligns rigid fragments, and hosts related modeling tools. ISOLDE couples interactive fitting to molecular mechanics, encouraging physically realistic rearrangements during model correction [12, 13]. Each has a different strength: interactive inspection is essential when automated software confidently places residues into incorrect density.

Interpretation warning

Do not automatically build chemistry into ambiguous blobs. A weak feature might represent a flexible loop, contaminant, alternate conformation, associated protein, nucleotide, lipid, ice artifact or noise. Assigning a ligand or DNA segment requires chemically and experimentally independent evidence, not simply a visually matching contour.

9 How to calculate a map from an atomic model

9.1 The forward operator for atomic refinement

Let atom j have position $\boldsymbol{r}_j$, occupancy o_j, and electron scattering factor $f_j^e(\boldsymbol{q})$ in the chosen approximation. An idealized atom-derived Fourier map coefficient is

$$F_c(\boldsymbol{q};\boldsymbol{\theta}) = \sum_{j=1}^{N} o_j f_j^e(\boldsymbol{q}) \exp\left(-\frac{B_j\|\boldsymbol{q}\|^2}{4}\right) \exp(-2\pi i \boldsymbol{q}\cdot\boldsymbol{r}_j). \tag{17}$$

The f_j^e term represents electron elastic scattering; the isotropic B_j term describes modelled attenuation associated with positional displacement or disorder. A model map $M_c = \mathcal{F}^{-1}\{F_c\}$ is then compared to a reconstructed experimental map using appropriate normalization, masking and resolution filtering. Real applications may need anisotropic displacement models, charge sensitivity, solvent corrections and additional details.

9.2 Convolution interpretation

Each atom contributes a localized shape g_j centered at $\boldsymbol{r}_j$:

$$M_c(\boldsymbol{r}) = \sum_{j=1}^{N} o_j g_j(\boldsymbol{r}-\boldsymbol{r}_j; B_j). \tag{18}$$

A sharpened experimental map and an artificially sharp calculated map are not necessarily comparable until common scale/resolution characteristics are considered. This is the main reason model-to-map fit scores must specify how the model map was calculated.

9.3 Occupancy versus displacement parameter

From Eq. (17), the amplitude of a contribution is approximately proportional to $o\exp[-Bs^2/4]$. A low occupancy o reduces amplitude at all frequencies; larger B reduces high-frequency amplitude more strongly. These parameters are partially confounded at limited resolution.

Worked example

Occupancy–B ambiguity. With $o = 0.5$, $B = 40\,\text{Å}^2$, and $s = 0.10\,\text{Å}^{-1}$, the relative amplitude is $0.5e^{-40(0.10)^2/4} = 0.452$. At $s = 0.25\,\text{Å}^{-1}$, it is $0.5e^{-40(0.25)^2/4} = 0.268$. A single low-frequency measurement cannot reliably separate occupancy and B; their differing frequency dependences can help if sufficiently good data span multiple frequencies.

9.4 Interpolation of experimental map values

Coordinates almost never fall exactly on voxel centers. Refinement therefore interpolates $M_o(\boldsymbol{r})$ and its gradients. In trilinear interpolation, if the fractional offsets within a unit grid cell are (u, v, w), then

$$M(u,v,w) = \sum_{a,b,c\in\{0,1\}} \eta_a(u)\eta_b(v)\eta_c(w)M_{abc}, \tag{19}$$

where $\eta_0(t) = 1 - t$ and $\eta_1(t) = t$. Other interpolation schemes may give smoother derivatives. Interpolation is an implementation detail with a direct impact on gradient-based refinement [10].

Worked example

Bilinear interpolation by hand. In a two-dimensional grid cell, the four corner values are $M_{00} = 2, M_{10} = 4, M_{01} = 6, M_{11} = 10$. At $u = 0.25, v = 0.60$,

$$M = (.75)(.40)(2) + (.25)(.40)(4) + (.75)(.60)(6) + (.25)(.60)(10) = 5.20.$$

The three-dimensional formula is the same product-weight logic with eight corners.

10 Real-space refinement as a Bayesian inverse problem

10.1 What exactly are we optimizing?

The unknown variables are the atomic coordinates, and possibly occupancies, displacement parameters, alternate conformers and domain transforms. Let $\boldsymbol{\theta}$ denote these parameters. An experimental map M_o provides a likelihood $p(M_o \mid \boldsymbol{\theta})$, while chemical information provides a prior $p(\boldsymbol{\theta})$. Bayes' rule gives

$$p(\boldsymbol{\theta} \mid M_o) \propto p(M_o \mid \boldsymbol{\theta})p(\boldsymbol{\theta}). \tag{20}$$

A maximum-a-posteriori estimate minimizes the negative log posterior:

$$\boxed{\widehat{\boldsymbol{\theta}} = \arg\min_{\boldsymbol{\theta}} \left[E_{\text{map}}(\boldsymbol{\theta}) + \lambda E_{\text{geometry}}(\boldsymbol{\theta})\right].} \tag{21}$$

Here λ is a relative weight. Phenix real-space refinement uses exactly this kind of combined map-and-restraint target, with algorithmic choices and extensions that differ from the illustrative forms below [10, 9].

10.2 A simple Gaussian map likelihood

If the map residuals are modeled as independent Gaussian errors (a simplifying assumption), then

$$E_{\text{map}}(\boldsymbol{\theta}) = \frac{1}{2\sigma^2} \sum_{v \in \Omega} [M_o(v) - M_c(v; \boldsymbol{\theta})]^2 . \tag{22}$$

In real reconstructions the noise is correlated, spatially varying and nontrivially filtered. More sophisticated targets can use normalized map correlation, spectral weighting and local noise models. The fundamental principle is unchanged: optimization must balance fit to data with prior chemical feasibility.

10.3 An alternative atom-centered target

A simplified local-fit reward is $T(\boldsymbol{\theta}) = -\sum_i w_i M_o(\boldsymbol{r}_i)$. This drives atoms toward higher map values. Its gradient with respect to coordinate $\boldsymbol{r}_i$ is

$$\boxed{\nabla_{\boldsymbol{r}_i} T = -w_i \nabla M_o(\boldsymbol{r}_i).} \tag{23}$$

This compact expression explains why interpolated map gradients are useful during coordinate optimization. It is only a didactic atom-centered objective; using it alone would move atoms into high contours without chemical context and may not reproduce a physically correct model [10].

10.4 Derive a harmonic-bond restraint

For a bond between atoms i, j with target distance d_0, a common penalty is

$$E_{\text{bond}} = \frac{k_b}{2} \left(\|\boldsymbol{r}_i - \boldsymbol{r}_j\| - d_0 \right)^2 . \tag{24}$$

Let $d = \|\boldsymbol{r}_i - \boldsymbol{r}_j\|$. Applying the chain rule,

$$\nabla_{\boldsymbol{r}_i} E_{\text{bond}} = k_b (d - d_0) \frac{\boldsymbol{r}_i - \boldsymbol{r}_j}{d}. \tag{25}$$

When the bond is too long, $d > d_0$, its gradient points outward and gradient descent moves the atom inward. This is the mathematical origin of chemically restrained model correction.

10.5 A hierarchy of geometric restraints

A useful total penalty can contain

$$E_{\text{geometry}} = E_{\text{bond}} + E_{\text{angle}} + E_{\text{dihedral}} + E_{\text{planar}} + E_{\text{chiral}} + E_{\text{clash}} + E_{\text{Rama}} + E_{\text{ref}}. \tag{26}$$

E_{ref} can constrain agreement with a trusted reference model, especially at low local resolution. Secondary-structure restraints and noncrystallographic symmetry information can reduce overfitting, but wrong annotations or incorrect reference structures can systematically bias the final coordinates [10, 9].

Worked example

Solve a refinement optimum exactly. Let x be a one-dimensional coordinate displacement. Set

$$E_{\text{map}} = (x - 0.65)^2, \qquad E_{\text{geom}} = 1.8(x + 0.30)^2.$$

Then $dE/dx = 2(x - 0.65) + 3.6(x + 0.30) = 0$. Hence $5.6x = 0.22$ and the optimum is $\boxed{x^* = 0.0393}$. The map-only solution is $x = 0.65$; the geometry-only solution is $x = -0.30$. The combined solution compromises between evidence and prior knowledge.

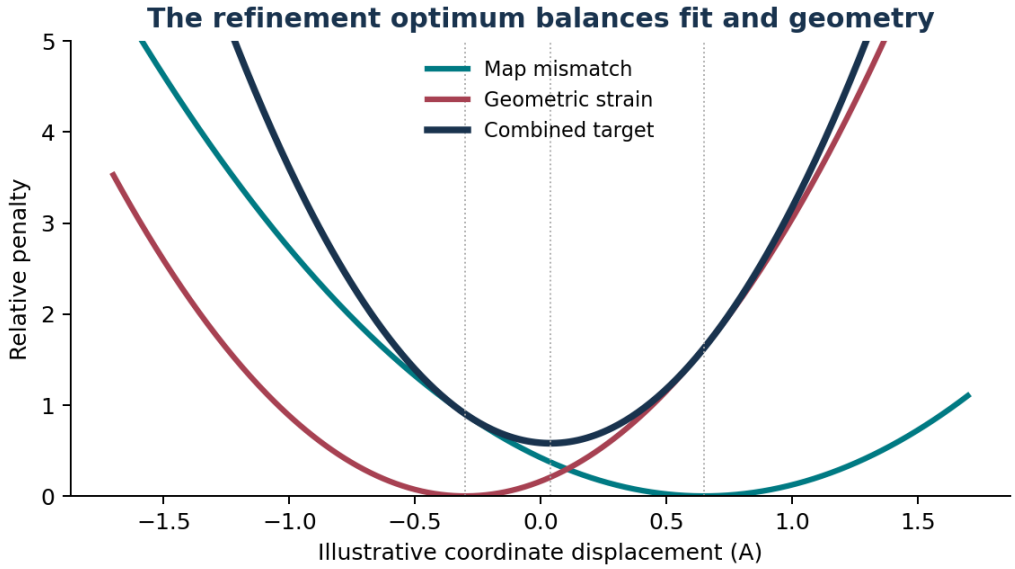


Figure 7: Synthetic one-coordinate optimization problem. A map-only optimum and a chemical-geometry optimum differ; the posterior/penalized optimum lies between them. The shape illustrates a mathematical tradeoff, not a measured protein force.

10.6 From maximum likelihood to Bayesian uncertainty

The optimal coordinate $\widehat{\boldsymbol{\theta}}$ is only a point estimate. Near a well-defined local optimum, a quadratic expansion of the negative log posterior gives

$$E(\boldsymbol{\theta}) \approx E(\widehat{\boldsymbol{\theta}}) + \tfrac{1}{2}(\boldsymbol{\theta} - \widehat{\boldsymbol{\theta}})^T H(\boldsymbol{\theta} - \widehat{\boldsymbol{\theta}}), \tag{27}$$

with Hessian $H = \nabla^2 E$. Under a Laplace/Gaussian approximation, the parameter covariance is H^{-1}, when invertible. Weak density, correlated parameters and restraints produce poorly constrained modes. Atomic models typically report coordinates, not a complete physically validated uncertainty distribution; interpret small structural changes accordingly.

11 Optimizing an atomic model in practice

11.1 Why refinement is iterative

Atomic-model refinement is nonconvex: moving a helix or flipping a side-chain rotamer may pass through worse intermediate fits before reaching a better solution. A common workflow alternates (i) model inspection and manual rebuilding, (ii) restrained automated refinement and (iii) validation. One should not trust the numerical optimizer to repair a grossly misplaced domain or incorrect sequence assignment without human intervention.

11.2 Coordinate descent, rigid groups and gradient descent

A simple gradient step is $\boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t - \eta_t \nabla E(\boldsymbol{\theta}_t)$. Practical implementations use optimized step selection, quasi-Newton methods, local searches, restraints, and sometimes simulated annealing. At modest resolution, start with 6-DOF rigid-body refinement of domains before refining all atomic coordinates. Adding many free parameters too early can overfit noise.

11.3 Why refinement weight selection matters

Write $E = E_{\text{map}} + \lambda E_{\text{restraint}}$. If λ is too small, the model follows noisy map fluctuations and develops distorted bonds or clashes. If too large, the model remains chemically neat but refuses real conformational changes. It

is not sound to choose λ solely to maximize final map-model CC, because that metric is not independent of the fitting objective. Phenix provides automatic restraints/map-weight strategies as well as model-validation outputs [9, 10].

11.4 Atomic displacement and occupancy refinement

B factors account for modeled signal attenuation from atomic disorder or displacement. Their refinement can be ill-conditioned if coordinates are incorrect or insufficient high-frequency information exists. Depending on software and protocol, coordinate and ADP refinements may occur in separate stages; the current Phenix reference notes specific implementation choices [9]. Occupancy refinement generally requires stronger information and constraints. A mobile loop with reduced mean density is not automatically a missing loop or a chemically unoccupied site.

11.5 Why simulated annealing can help but not certify accuracy

Simulated annealing explores conformational changes by temporarily permitting larger displacements and then cooling toward low-objective regions. It can escape some local minima. But the result still depends on map quality, restraints and initial state. A low objective does not certify correct sequence registration or biological relevance.

12 Quantifying model-to-map agreement

12.1 Normalized map correlation

For map values m_v and calculated model values c_v within a specified voxel set Ω, the Pearson-type correlation is

$$\mathrm{CC}(m,c) = \frac{\sum_{v\in\Omega}(m_v-\bar{m})(c_v-\bar{c})}{\sqrt{\sum_{v\in\Omega}(m_v-\bar{m})^2\sum_{v\in\Omega}(c_v-\bar{c})^2}}. \tag{28}$$

Map CC depends on the comparison region, resolution, filtering, map normalization and the constructed model map. A favorable whole-map CC can conceal a badly modeled loop. Report local or per-residue diagnostic metrics as well [9, 19].

12.2 Map-model Fourier-shell correlation

Generate $F_c(\boldsymbol{q})$ from the atomic model and compare against an experimental Fourier map $F_o(\boldsymbol{q})$ in shells, analogous to Eq. (2):

$$\mathrm{FSC}_{\mathrm{model,map}}(q) = \frac{\sum_{\Omega_q} F_o F_c^*}{\sqrt{\sum_{\Omega_q}|F_o|^2\sum_{\Omega_q}|F_c|^2}}. \tag{29}$$

This *is not* the same as half-map FSC. Half-map FSC compares two independent data reconstructions. Model-map FSC compares a fitted structural hypothesis to a reconstructed map. They answer different questions and need not cross any one threshold at the same frequency [19].

12.3 Per-atom Q-scores

The Q-score introduced by Pintilie and colleagues measures how the local map profile around an atom resembles an idealized resolvable atom profile, typically using correlations of radial map profiles. A high atom Q-score generally indicates strong local resolvability assuming the atom is correctly placed; it does not independently establish chemical identity. Per-residue averages can locate regions for manual examination [15].

12.4 Side-chain validation with EMRinger

EMRinger tests whether the density sampled while rotating candidate side-chain chi angles peaks near chemically favored rotamer positions, evaluating local map support for side-chain orientations [16]. It complements all-atom steric validation; a well-placed helix can have poor side-chain register, and a chemically allowed rotamer can lack convincing experimental support.

12.5 Difference maps: where model and experiment disagree

A simplified real-space residual map is

$$D(\boldsymbol{r}) = M_o(\boldsymbol{r}) - \alpha M_c(\boldsymbol{r}), \tag{30}$$

where α adjusts the relative scale. Positive features may indicate unmodeled signal; negative features may indicate excess calculated model contribution. The result is only interpretable if map/model scaling, B factors, resolution and local uncertainty are handled consistently. Servalcat provides a more sophisticated approach to map calculation and refinement [14].

Worked example

Why whole-map CC can conceal a bad loop. Suppose 900 voxels in a rigid core are predicted extremely well, while 100 voxels around a loop are predicted poorly. A global correlation computed over all 1000 may remain high because the rigid core dominates its variance. Recompute CC within a scientifically justified mask around the loop, and examine the local half-map evidence. The global number is not an adequate test of the loop conformation.

13 Stereochemical model validation

13.1 Why chemistry is independent evidence

The map supplies structural evidence, but stereochemistry supplies a second set of constraints. A model with impressive map correlation can still have impossible bond lengths, wrong chirality or steric clashes. MolProbity and related tools assess bond/angle deviations, Ramachandran outliers, rotamers, all-atom contacts, C-beta deviations, peptide geometry and nucleic-acid-specific parameters [17, 18].

13.2 Clashscore

A steric overlap is diagnosed when nonbonded atoms come implausibly close after accounting for atomic radii and chemistry. A commonly used clashscore is the number of serious clashes per 1000 atoms:

$$\text{Clashscore} = 1000 \frac{N_{\text{serious clashes}}}{N_{\text{atoms}}}. \tag{31}$$

Its exact clash definition is software-specific (including hydrogen addition and overlap threshold). One should not apply an arbitrary cutoff without consulting the method used by the validation program.

13.3 Ramachandran and rotamer outliers

For each backbone residue, compute ϕ, ψ and compare with empirical residue-class distributions. For a side chain, compute χ torsions and compare with rotamer libraries. Outliers are inspection targets, not automatic deletion targets: strained catalytic or bound conformations may be correct, but require unusually clear evidence. An outlier in a weak density region is more suspicious than one supported by high local resolution and specific contacts.

Worked example

Clashscore arithmetic. If a model contains 8,200 atoms and 13 serious clashes under the chosen validator definition, its clashscore is $1000 \times 13/8200 = 1.59$. This is an easy arithmetic score, but whether the clashes occur in the substrate-binding site or in low-confidence surface loops is often more biologically consequential than the global average.

14 Cross-validation and the danger of overfitting

14.1 Training fit versus independent prediction

If an atomic model is refined against the full reconstructed map, its map CC is a measure of *training agreement*. Repeatedly adjusting thousands of coordinates can exploit noise and correlated map errors. A robust evaluation asks whether structural features are supported in independent data, not merely whether they fit the optimization target.

14.2 Using half maps to assess overfitting

One possible validation scheme refines against a designated working half map and evaluates map-model agreement against the independent other half map, with appropriate filtering and harmonized map processing. The difference between model–work and model–test correlations can reveal overfitting. However, if the model was initially built using the full map or if the two maps share information through processing, the test is not truly independent. Thus, *hold-out claims must describe exactly what the model has seen*. Half-map cross-validation is related to, but not identical with, crystallographic R_{free} [19, 20].

14.3 Map–model alignment and resolution leakage

Modeling against an aggressively sharpened or heavily masked map may optimize artificial features. A density-modified or machine-enhanced map may incorporate external structural priors. Such maps can assist interpretation, but a final structural claim should be traceable to original reconstruction-derived evidence. Global half-map FSC itself does not certify a particular ligand or side-chain placement.

Interpretation warning

Common circular argument. "The model fits the map; therefore the model is correct." But if the model was used to fit, sharpen, enhance, segment or create that map, fit is partially guaranteed by construction. Report independent data checks, local map quality, geometry and competing interpretations.

15 What can and cannot be inferred from a single map?

15.1 The map may be an average over states

A reconstruction of several conformations can be written schematically

$$M(\boldsymbol{r}) = \sum_{k=1}^{K} \pi_k \big(h_k * V_k\big)(\boldsymbol{r}) + \epsilon(\boldsymbol{r}), \qquad \pi_k \geq 0, \quad \sum_k \pi_k = 1. \tag{32}$$

A blurred region might reflect motion, partial occupancy, conformational mixing, local noise or a combination. Refining a *single* atomic coordinate model into that average can yield an artificial structure not substantially occupied by any particle.

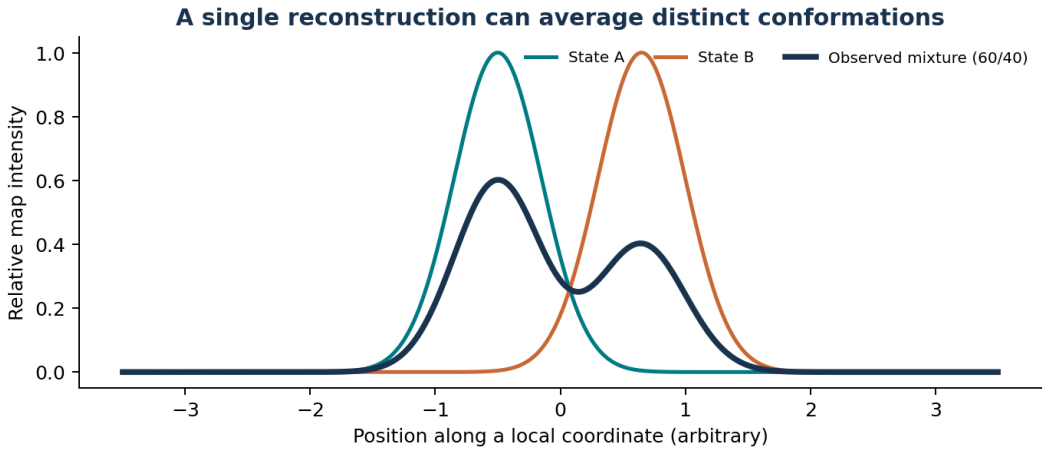


Figure 8: A conceptual one-dimensional mixture. An ensemble average need not correspond to a single physical conformation. A single-coordinate refinement of the mixture can misrepresent both components.

15.2 A refined coordinate difference is not automatically a force measurement

Suppose two atomic models differ in a local bond angle or domain separation. This directly supports a *structural difference* only to the precision permitted by the maps and controls. Converting it to a physical force requires a calibrated mechanical model (or appropriate independent force measurement), not merely Hooke's-law analogy. Thermal fluctuations, linker compliance, solvent and state selection may alter the relationship between imposed condition and inferred geometry.

15.3 AlphaFold predictions are priors, not cryo-EM observations

A predicted model is useful for selecting secondary structures, reference geometry and plausible fold architecture. But using it as a reference restraint makes the refined model partially prediction-dependent. Difference claims should report how strongly the answer changes when the reference model is relaxed or replaced. A high model confidence score cannot substitute for independent map support.

16 Worked interpretation for a PP2A A-subunit DNA-spring chimera

16.1 Why this is an especially difficult case

Your PP2A A-subunit construct is an elongated, repeat-rich scaffold with 15 HEAT repeats. In the current experimental design, tetrazine substitutions near D6 and N552 are joined by a DNA-spring construct, presently using approximately 80-nt DNA. The cryo-EM problem is not simply to fit an average PP2A structure: one must separate particle alignment error, intrinsic flexibility, DNA-induced contacts, genuine mechanical bias, and biochemical differences among samples. The supplied 2025 *Structure* paper on PP2A A-subunit isoforms discusses conformational differences within assembled complexes, supporting flexibility of the scaffold without by itself demonstrating an imposed DNA-spring force effect [21]. Likewise, the 2026 ubiquitin-DNA preprint supplies a cautionary control example: much of the conformational effect was reproduced by DNA attachment rather than duplex spring tension [22].

16.2 Recommended order of interpretation

First compare the relaxed and hybridized constructs in matched chemistry and sequence backgrounds. Reconstruct with comparable masks, filtering and sampling. Verify global and local resolution, directional coverage and half-map FSC independently. Align the stable shared HEAT-repeat core before comparing the ends. If one region appears displaced, test sensitivity to which residues are used for alignment and to distinct

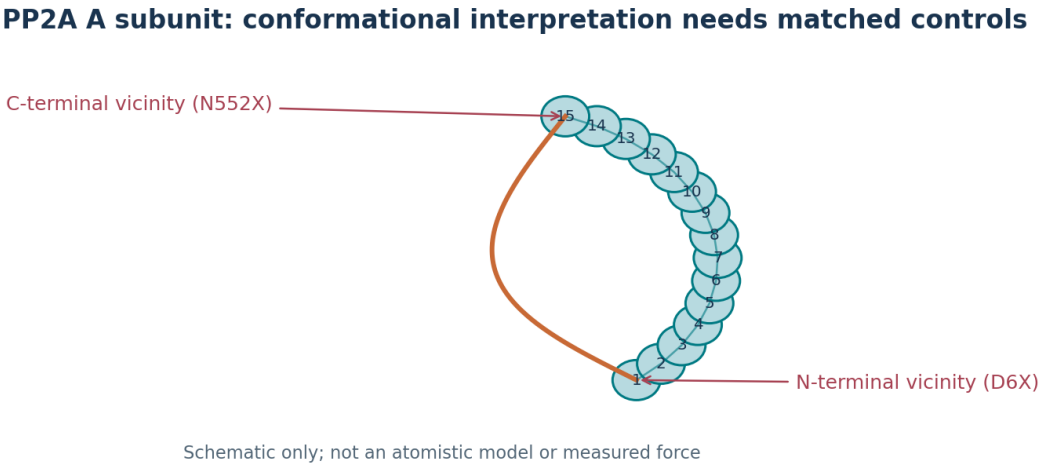


Figure 9: Original conceptual geometry of the 15-repeat PP2A A subunit, with an external DNA segment linking N- and C-terminal regions. The drawing is not an experimentally refined atomic model and does not imply a calibrated force.

starting model priors. A nicked/gapped duplex control can distinguish DNA proximity from duplex mechanical stress.

16.3 Domain-level structural descriptors

For each repeat k, define a reference domain frame $(R_k, \boldsymbol{t}_k)$, or a center-of-mass coordinate $\boldsymbol{c}_k$. A global bending descriptor can be expressed as a change in angle between selected domain vectors:

$$\theta = \arccos \frac{(\boldsymbol{c}_b - \boldsymbol{c}_a) \cdot (\boldsymbol{c}_d - \boldsymbol{c}_c)}{\|\boldsymbol{c}_b - \boldsymbol{c}_a\| \, \|\boldsymbol{c}_d - \boldsymbol{c}_c\|}. \tag{33}$$

Compare $\Delta\theta = \theta_{\text{spring}} - \theta_{\text{control}}$ only with uncertainty estimates. Inter-repeat rigid-body transformations are often better-identified than per-atom dihedral changes at moderate local resolution.

16.4 RMSD and alignment dependence

A coordinate RMSD requires choosing corresponding atoms and an alignment region. If you superpose *all* repeats, the fitted rotation may partially erase a real terminal displacement. If instead you align a stable central region, terminal motions remain visible, but the resulting RMSD includes uncertainties in that reference frame. Report the atom selection, state assignments and map support along with RMSD.

16.5 The danger of inferring DNA spring shape from blurred density

An 80-bp double helix can be bent, disordered or only partly resolved. If its density is weak, an apparently smooth fitted duplex may simply be a geometry prior. Do not quote an exact end-to-end force, kinking angle or local base-pair disruption unless the data independently constrain the DNA geometry and a validated mechanical relationship maps that geometry to load.

Key concept

Most defensible hierarchy for the PP2A project. (i) Is there reproducible map-level redistribution between matched conditions? (ii) Is there a statistically supported shift in domain geometry? (iii) Can backbone/side-chain changes be resolved locally? (iv) Do assembly or phosphatase assays respond? (v) Can a calibrated spring model translate the perturbation into force? Each step requires additional evidence.

17 An end-to-end operational workflow

17.1 The map stage: CryoSPARC

Start from validated 3D refinement or class-specific maps and the corresponding particles. Preserve the half maps. Use gold-standard FSC and inspect the refinement masks. Estimate local resolution and directional anisotropy. For flexible regions, consider local refinement with a soft mask and, if justified, particle subtraction. Generate separate display-oriented sharpened/locally filtered maps without replacing raw evidence [6, 3, 1, 5, 7].

17.2 The coordinate stage: ChimeraX, Coot or ModelAngelo

Inspect maps at several contour thresholds. Verify pixel size, handedness, molecular identity and biological assembly. Place an existing model rigidly where appropriate, preferably by trustworthy domains. Trace missing chains only where the map supports them. For high-quality maps, ModelAngelo can provide an initial atomic model; Coot and ChimeraX support inspection, manual rebuilding and fragment adjustments [11, 12].

17.3 The refinement stage: Phenix, ISOLDE or Servalcat

Choose a map suitable for refinement, preserve the independent half maps, and adjust map-versus-geometry weights. Correct domain misplacement before moving all atom coordinates. Iterate stereochemical corrections, local map inspection and restrained refinement. For large misfits or strained regions, physics-guided interactive remodeling can help, but must be validated against the data [9, 10, 13, 14].

17.4 Validation and reporting

Archive the final primary map, native half maps, masks, FSC curves, resolution statistics, coordinate model, starting-model provenance and refinement logs. Report map-model FSC, local fit and geometry diagnostics, explain unresolved regions, identify unsupported ligands, and record deposited PDB/mmCIF and EMDB accession information. Current EMDB policy requires original half maps for SPA/STA/helical structures when the processing pipeline produces them; deposited unfiltered half maps must not be postprocessed after refinement [20, 19].

Deliverable	**What to check before release**
Primary map and half maps	Same coordinates, voxel sizes, handedness; half maps retained unfiltered
FSC and masks	Correct half sets, soft masks, transparent threshold and local-resolution claims
Atomic model (mmCIF/PDB)	Correct sequence registration, molecules, chain IDs, ligands, occupancies
Chemical geometry	Bond/angle deviations, clashes, Ramachandran, rotamers, peptide geometry
Experimental model support	Map-model FSC, residue-wise fit, competing states and independent checks
Methods and uncertainty	Starting models, software settings, masks, local resolution and limitations

18 Exercises

The exercises are designed to be answerable from the chapter alone. Unless stated otherwise, use the simplified models introduced above. Many real-world quantities require experimentally estimated noise or calibrated algorithms and therefore do not have universal numerical cutoffs.

1. **Sampling and resolution.** A map grid has $a = 1.5\,\text{Å}$ voxels and a half-map FSC crossing at $d = 4.8\,\text{Å}$. Compute the Nyquist spacing and explain which quantity limits structural interpretation.

2. **Sharpening gain.** At $s = 0.20\,\text{Å}^{-1}$, calculate the multiplier for $B_{\text{sharp}} = -60\,\text{Å}^2$. State the effect on noise.

3. **Half-map SNR.** For $r = 0.4$ half-map FSC, compute $\mathrm{SNR}_{\mathrm{half}}$ and the equal-noise full-map correlation.

4. **Rigid-body alignment.** A model is translated by $\boldsymbol{t} = (2, -3, 5)$ Å and rotated by 90° about the z-axis. Find the target position of $\boldsymbol{x} = (1, 0, 0)$ Å.

5. **Bond restraint.** With $d = 1.60$ Å, target $d_0 = 1.50$ Å and $k_b = 100$ in arbitrary units, evaluate E_{bond} and the gradient magnitude on one atom.

6. **Occupancy/B effects.** If o drops from 1 to 0.5 at fixed B, what happens to the modeled amplitude at all s? How does increasing B differ?

7. **Model uncertainty.** A loop is locally 7 Å while the core is locally 3 Å. Which region supports individual side-chain rotamer claims more strongly? Why?

8. **Model selection.** An atomic refinement improves global map CC from 0.82 to 0.86 but raises the clashscore from 3 to 24. What should the researcher do next?

9. **Validation leakage.** A model was built using the full map, then refined against one half-map, and tested on the other. Why is this not a fully untouched validation set?

10. **Biological inference.** Two PP2A spring states show a 3° difference in a domain angle. State at least four independent analyses needed before claiming a mechanistically meaningful spring response.

19 Solutions and explanations

1. $d_N = 2a = 3.0$ Å. The map can represent features down to the sampling bound in principle, but the measured 4.8 Å half-map resolution is the more conservative global evidence limit, with local and directional limitations still possible.

2. $H_B = e^{-(-60)(0.20)^2/4} = e^{0.6} = 1.82$. It multiplies both signal and noise at that frequency by 1.82; it does not increase the intrinsic SNR.

3. $\mathrm{SNR}_{\mathrm{half}} = r/(1-r) = 0.4/0.6 = 2/3$. The equal-noise combined-map FSC is $2(0.4)/(1+0.4) = 0.571$.

4. A 90° z-rotation maps $(1, 0, 0)$ to $(0, 1, 0)$; adding $\boldsymbol{t}$ gives $(2, -2, 5)$ Å.

5. $E = \frac{1}{2}100(0.1)^2 = 0.5$ arbitrary energy units. The gradient magnitude is $k_b|d - d_0| = 10$ arbitrary energy units per angstrom; gradient descent tends to shorten the stretched bond.

6. Halving occupancy halves the calculated amplitude independently of s, in the model. Increasing B applies a stronger factor $e^{-Bs^2/4}$ at larger frequencies, weakening fine detail disproportionately.

7. The 3 Å core usually provides stronger side-chain evidence, subject to local map details, composition and validation. A 7 Å loop may constrain its envelope but not individual rotamers or hydrogen-bond geometry.

8. Inspect the newly introduced clashes and map support, evaluate map-vs-restraint weighting, compare held-out correlations, and rebuild problematic regions. Maximizing global CC while destroying chemistry is not a successful refinement.

9. Both half-maps contributed to the full map used in building the initial model, so the alleged held-out map influenced model selection. Subsequent testing is partly contaminated; a fully independent claim requires more careful data provenance and assessment.

10. At minimum: compare maps with matched filtering and masks; quantify local/directional resolution in the relevant region; test domain-angle stability under different alignment masks and independent refinements; propagate conformational uncertainty across half maps or resamples; and verify matched chemical controls (relaxed/nicked/duplex) as well as biochemical assays. A 3° angle difference alone does not determine force or activity.

20 Final synthesis: what should be called "complete"?

There is no universal last button in cryo-EM. A convincing endpoint is a *supported scientific claim*: independent map reconstruction establishes the signal, regional resolution identifies what can be interpreted, atomic coordinates (if justified) explain the map without pathological chemistry, independent validation avoids circular fitting arguments, and the reported biological interpretation does not exceed the experiment's controls or precision. A map-only result can be complete; conversely, an elaborate atomic model is incomplete if its crucial residues are unsupported.

Key concept

The three-chapter logic. (I) Electron–specimen interactions and microscope optics determine the noisy particle images. (II) Reconstruction and classification use many such images to infer three-dimensional structure and heterogeneity. (III) Map postprocessing, atomistic model building and validation turn those reconstructions into constrained structural statements. Every downstream conclusion inherits uncertainty from the preceding stages.

Selected literature and documentation

The references below were chosen to distinguish primary methodological research, current software documentation and community validation standards. Online documentation may change with software releases; links were checked or located on 28 October 2026. All figures in this chapter are original schematic or synthetic illustrations, not reproduced experimental micrographs.

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